Chapter 6F

Endocrine Review - Integrated Cases and Algorithms

Original Medvale study chapter. Designed for concept-rich reading, exam synthesis, and future graph-RAG extraction. Source uploads were used as topic coverage signals; wording, tables, algorithms, and cases are original.

Clinical purpose

Endocrinology is a pattern-recognition specialty, but the patterns become reliable only when the learner anchors them to feedback loops, hormone source, target-organ response, and time course. The same patient may have a biochemical abnormality, a structural lesion, an acute endocrine emergency, and a chronic risk-modification problem at the same time. This final endocrine segment ties together diabetes, thyroid disease, adrenal disease, pituitary disease, diabetes insipidus, SIADH, and selected calcium-bone themes into a practical diagnostic framework.

A useful endocrine question has four parts: Which axis is abnormal? Is the hormone excessive or deficient? Is the gland producing the abnormality primary to the target gland or secondary to central control? Is the patient stable enough for confirmatory testing, or must treatment start immediately?

Core emergency principle: DKA/HHS, severe hypoglycemia, thyroid storm, myxedema coma, adrenal crisis, severe symptomatic hyponatremia, and severe hypercalcemia should be treated while confirmatory details are being gathered.

One-page endocrine operating system

Syndrome-first triage map

Hyperglycemic crisis	Ketosis/acidosis vs hyperosmolarity	Glucose, beta-hydroxybutyrate/ketones, BMP, pH, anion gap, serum osmolality	Cerebral edema, hypokalemia during insulin, shock, infection	Starting insulin before potassium; missing euglycemic DKA
Thyrotoxicosis	True hyperthyroidism vs destructive release or exogenous hormone	TSH, free T4, T3, antibodies, uptake when appropriate	AF, heart failure, thyroid storm	Treating all low-TSH states like Graves
Hypothyroidism	Primary thyroid failure vs central disease	TSH and free T4; pituitary workup if TSH not appropriate	Myxedema coma, hypothermia, hypoventilation, hyponatremia	Using TSH alone when pituitary disease is possible
Cushing phenotype	Exogenous steroid vs endogenous cortisol excess	Medication history, late-night salivary cortisol, dex suppression, urine cortisol	Infection, VTE, severe HTN, diabetes, psychosis	Testing low-pretest-probability patients during acute stress
Pheochromocytoma	Catecholamine spells vs common HTN/anxiety	Plasma free or urine fractionated metanephrines	Hypertensive crisis, arrhythmia, stroke	Giving beta blocker before alpha blocker
Primary aldosteronism	Renin-suppressed HTN with hypokalemia/resistant HTN	Aldosterone-renin ratio, confirmatory/localization testing	Stroke, CKD, AF, refractory HTN	Assuming normokalemia excludes it
Pituitary mass	Hormone excess/deficiency vs mass effect	MRI, prolactin, IGF-1, cortisol/ACTH, TSH/free T4, gonadotropins	Optic chiasm compression, adrenal insufficiency	Confusing stalk effect with prolactinoma
Polyuria-polydipsia	Solute diuresis vs water diuresis	Glucose, Ca, K, serum Na/osm, urine osm	Hypernatremia/dehydrati if no water access	Calling all polyuria DI before excluding DM

Diagnostic algorithms

Algorithm 1: Hyperglycemic crisis at the bedside

- Assess airway, mental status, perfusion, volume status, infection/MI/stroke clues.
- Check glucose, potassium, sodium, bicarbonate, anion gap, creatinine, beta-hydroxybutyrate or ketones, venous/arterial pH, and serum osmolality.
- Start isotonic fluids early unless there is a compelling contraindication.
- If potassium is below 3.3 mEq/L, hold insulin and replete potassium first.
- If potassium is safe, start insulin; monitor glucose, potassium, anion gap, and mental status.
- Find and treat the trigger: infection, missed insulin, MI, stroke, surgery, pregnancy, medication, substance use, or endocrine stressor.
- Transition to subcutaneous insulin only when clinically improved and DKA gap closure or HHS osmolar improvement has occurred.

DKA is insulin deficiency plus counterregulatory hormone excess. The sequence is lipolysis, ketogenesis, anion-gap metabolic acidosis, osmotic diuresis, and volume depletion. HHS is dominated by severe hyperglycemia, hyperosmolality, and dehydration, usually with enough insulin activity to restrain ketogenesis. Both syndromes are potassium-depleted states even when the first serum potassium is normal or high.

DKA versus HHS

Typical setting	Type 1 DM or insulin-deficient type 2 DM; SGLT2 inhibitors can cause euglycemic DKA	Older type 2 DM, infection, limited water access, stroke, MI, steroids	Age and diabetes type do not absolutely rule in/out either syndrome
Glucose	Often >250 mg/dL but may be lower in euglycemic DKA	Often >600 mg/dL	Glucose alone is not the diagnosis
Ketones	Prominent beta-hydroxybutyrate/acetoacetate	Absent or mild	Urine ketones can underestimate beta-hydroxybutyrate
pH/bicarbonate	Acidosis, low bicarbonate	pH often >7.30 and bicarbonate >18 unless mixed disorder	Acid-base status guides monitoring intensity
Osmolality	Elevated variably	Markedly elevated, often >320 mOsm/kg	Mental status tracks osmolality more than glucose alone

Algorithm 2: Low TSH

- Confirm free T4 and T3. Low TSH with high T4/T3 equals thyrotoxicosis.
- Look for Graves signs: diffuse goiter, thyroid bruit, orbitopathy, pretibial myxedema.
- Look for thyroiditis signs: painful thyroid, viral prodrome, postpartum state, low uptake.
- If not pregnant and diagnosis is unclear, uptake pattern helps: diffuse uptake suggests Graves; focal or multinodular uptake suggests toxic adenoma/multinodular goiter; low uptake suggests thyroiditis, exogenous hormone, or iodine effect.
- Treat beta-adrenergic symptoms early; choose thionamide, radioiodine, surgery, or supportive care based on mechanism.

A low TSH is not automatically Graves disease. Graves disease is hormone overproduction driven by TSH-receptor stimulation. Toxic adenoma or toxic multinodular goiter is autonomous hormone production. Thyroiditis releases preformed hormone from damaged follicles and usually has low uptake. Exogenous hormone also suppresses uptake.

Algorithm 3: High TSH or suspected hypothyroidism

- Check free T4. High TSH + low free T4 is overt primary hypothyroidism.
- High TSH + normal free T4 is subclinical hypothyroidism; interpret with symptoms, pregnancy, TPO antibodies, goiter, LDL/CV risk, and TSH magnitude.
- Low free T4 with low/normal/inappropriately mild TSH suggests central hypothyroidism.
- Treat overt primary hypothyroidism with levothyroxine, starting lower in older patients or CAD.
- If central hypothyroidism is possible, evaluate adrenal function before thyroid hormone replacement.

Central hypothyroidism is a pituitary/hypothalamic problem. TSH may be low, normal, or slightly high but biologically ineffective. In pituitary disease, the free T4 is the diagnostic anchor, not the TSH alone.

Algorithm 4: Suspected adrenal insufficiency

- If the patient is in shock, severely ill, or has adrenal crisis features: draw cortisol/ACTH if this does not delay therapy, then give stress-dose hydrocortisone immediately.
- If stable: obtain morning cortisol and ACTH; use cosyntropin stimulation when needed.
- Low cortisol + high ACTH suggests primary adrenal insufficiency.
- Low cortisol + low or inappropriately normal ACTH suggests central adrenal insufficiency.

• Primary adrenal insufficiency often requires both glucocorticoid and mineralocorticoid replacement; cent usually preserves aldosterone.

Primary adrenal insufficiency damages the adrenal cortex itself, causing cortisol and aldosterone deficiency with elevated ACTH. Hyperpigmentation, salt craving, hypotension, hyponatremia, and hyperkalemia point toward primary disease. Secondary or tertiary disease often follows chronic glucocorticoid exposure or pituitary disease and usually does not cause hyperkalemia.

Algorithm 5: Adrenal hypertension

- Resistant hypertension or hypertension with hypokalemia: screen for primary aldosteronism with aldosterone-renin ratio.
- Paroxysmal hypertension with headache, sweating, and palpitations: screen for pheochromocytoma/paraganglioma with plasma free or urine fractionated metanephrines.
- Cushing phenotype with proximal weakness, bruising, striae, osteoporosis, diabetes, or infections: screen for cortisol excess.
- Localize adrenal disease after biochemical confirmation whenever feasible because incidental adrenal nodules are common.
- For pheochromocytoma, alpha blockade comes before beta blockade.

Master comparison tables

Diabetes management and complications

Type 1 DM	Autoimmune beta-cell destruction; absolute insulin deficiency	Hyperglycemia with symptoms, A1c/glucose criteria; antibodies/C-peptide when uncertain	Lifelong insulin, basal-bolus education, sick-day rules, ketone awareness	Young patient wit weight loss, keto DKA
Type 2 DM	Insulin resistance plus progressive beta-cell dysfunction	A1c >=6.5%, fasting glucose >=126, 2-h OGTT >=200, or random glucose >=200 with symptoms	Lifestyle, metformin when appropriate, GLP-1 RA/SGLT2i by ASCVD/HF/CKD/weight, insulin when catabolic or severe	Obesity, acanthos hypertriglyceride liver
Diabetic kidney disease	Hyperfiltration, albuminuria, mesangial expansion, fibrosis	Urine albumin-creatinine ratio and eGFR	ACEi/ARB for albuminuria/HTN, SGLT2i when eligible, BP/glycemic control	Albuminuria can p creatinine rise
Retinopathy	Microvascular leakage and ischemia leading to neovascularization	Dilated eye exams by diabetes type/duration	Glycemic and BP control; ophthalmology; anti-VEGF/laser when indicated	Floaters/vision l vitreous hemorrha
Neuropathy/foot	Axonal injury, ischemia, glycation, impaired wound healing	Monofilament, vibration, pulses, foot inspection	Foot care, off-loading, infection evaluation, pain meds	Painless ulcer ur metatarsal head; glove symptoms

Thyroid disease pattern table

Graves	Low TSH, high T4/T3	Diffuse goiter, orbitopathy, tremor, heat intolerance, tachycardia	TSH receptor antibody; diffuse uptake	Beta-blocker; methimazole i nonpregnant adults; radioiodine/surgery selecte
Toxic nodule/multinodular goiter	Low TSH, high hormone	Older patient, nodular asymmetric gland	Focal or patchy uptake	Radioiodine or surgery; bet stabilization
Subacute thyroiditis	Low TSH early, later high TSH possible	Painful tender thyroid after viral illness; elevated ESR	Low uptake despite thyrotoxicosis	NSAIDs/steroids for pain; b blocker; no thionamide mech
Hashimoto hypothyroidism	High TSH, low/normal free T4	Fatigue, cold intolerance, constipation, bradycardia, goiter	TPO antibodies	Levothyroxine for overt dis monitor TSH in primary dise
Thyroid storm	Thyrotoxicosis plus systemic collapse	Fever, delirium, tachyarrhythmia, HF, GI symptoms	Clinical emergency	Beta-blocker, thionamide, i thionamide, glucocorticoid,
Myxedema coma	Severe low thyroid hormone	Hypothermia, bradycardia, hypoventilation, altered mental status	Clinical emergency	IV thyroid hormone, stress- steroids, warming/ventilati precipitant

Adrenal and pituitary hormone excess/deficiency

Cushing syndrome	Cortisol excess	Screen with late-night salivary cortisol, dex suppression, or 24-h urinary free cortisol; then ACTH-dependent vs independent	Proximal weakness, easy bruising, striae, osteoporosis, diabetes, infection	Treat source; steroids; sur
Pheochromocytoma	Catecholamine excess	Metanephrines first; image after biochemical evidence	Paroxysmal headache, sweating, palpitations, resistant HTN; MEN2	Alpha blockad blockade; sur
Primary aldosteronism	Aldosterone excess with suppressed renin	Aldosterone-renin ratio; confirm/localize as indicated	Resistant HTN, hypokalemia, metabolic alkalosis; can be normokalemic	Adrenalectomy adenoma; MR a bilateral dis
Primary adrenal insufficiency	Cortisol and aldosterone deficiency; high ACTH	Morning cortisol/ACTH and cosyntropin; treat crisis immediately	Hyperpigmentation, hypotension, hyponatremia, hyperkalemia	Hydrocortison fludrocortiso education
Prolactinoma	Prolactin excess	Serum prolactin; exclude pregnancy, hypothyroidism, renal failure, drugs; MRI	Amenorrhea/galactorrhea, infertility, hypogonadism, visual deficits	Dopamine agon selected refr cases
Acromegaly	GH/IGF-1 excess	IGF-1 screening; oral glucose fails to suppress GH; MRI	Enlarged hands/feet, coarse features, OSA, cardiomyopathy, diabetes, colon polyps	Transsphenoid somatostatin if persistent
Hypopituitarism	Loss of pituitary hormones	Low target hormones with low/inappropriately normal trophic hormones; MRI	Fatigue, amenorrhea/low libido, adrenal insufficiency, central hypothyroidism	Replace corti hormone; repl indicated

Water balance, sodium, and pituitary links

Endocrine water disorders are missed because the complaint may be nonspecific: polyuria, thirst, dizziness, confusion, weakness, or a sodium abnormality. The diagnostic split is between water retention, free-water loss, and solute-driven water loss. SIADH causes water retention with inappropriately concentrated urine and euvolemic hyponatremia. Diabetes insipidus causes water diuresis, dilute urine, thirst, and hypernatremia risk. Diabetes mellitus causes solute diuresis from glucosuria, which can mimic DI clinically but has a different mechanism and treatment.

DI, primary polydipsia, and SIADH

Core defect	Low ADH secretion	Kidney resistance to ADH	Excess water intake suppresses ADH	Excess ADH effect des tonicity
Serum Na/osm	Normal-high or high if no water access	Normal-high or high	Low-normal or low	Low sodium, low serum
Urine osm	Low; rises after desmopressin	Low; little/no response to desmopressin	Low initially; rises with water deprivation	Inappropriately high hypotonic serum
Causes	Pituitary surgery, trauma, tumor, infiltrative disease, idiopathic	Lithium, hypercalcemia, hypokalemia, tubulointerstitial disease	Psychiatric disease, habit, thirst disorder	Pulmonary/CNS disease malignancy, drugs, po pain/nausea
Treatment	Desmopressin and water access	Treat cause; low solute diet, thiazide selected cases	Behavioral/psychiatric management; avoid rapid correction	Fluid restriction; salt/loop/urea/vaptar hypertonic saline if symptoms

Hyponatremia trap: Correct tonicity first. Hyperglycemia causes translocational hyponatremia. Pseudohyponatremia from marked lipid/protein elevation lowers measured sodium without true hypotonicity. True hypotonic hyponatremia is then sorted by urine osmolality, urine sodium, and effective arterial blood volume.

Integrated cases

Case 1 - Vomiting, abdominal pain, and altered breathing

Vignette: A 19-year-old with weight loss, polyuria, and vomiting has glucose 480 mg/dL, bicarbonate 9 mEq/L, anion gap 25, positive beta-hydroxybutyrate, potassium 5.4 mEq/L, and Kussmaul respirations.

Interpretation: DKA. High serum potassium does not mean adequate total-body potassium. Start isotonic fluids, monitor/replete potassium, start insulin when safe, find the trigger, and continue insulin until the anion gap closes.

Case 2 - Severe hyperglycemia but no ketones

Vignette: A 78-year-old with pneumonia is confused and profoundly dehydrated. Glucose is 960 mg/dL, serum osmolality is 342 mOsm/kg, bicarbonate is 22 mEq/L, and ketones are absent.

Interpretation: HHS. The dominant danger is hyperosmolality and volume depletion. Treat with aggressive fluids, electrolyte monitoring, low-dose insulin after initial resuscitation, and infection control.

Case 3 - Low TSH after a painful URI

Vignette: A patient has palpitations, neck pain, feverish prodrome, low TSH, high free T4, and a tender thyroid. Uptake scan is low.

Interpretation: Subacute thyroiditis. Hormone is leaking from inflamed follicles, so thionamides are not the main therapy. Use beta-blockers and NSAIDs or glucocorticoids for pain/inflammation.

Case 4 - Low TSH, eye symptoms, and diffuse goiter

Vignette: A patient has tremor, weight loss, heat intolerance, diffuse goiter with bruit, proptosis, low TSH, and high T3/T4.

Interpretation: Graves disease. Use beta-blocker for symptoms and choose antithyroid medication, radioiodine, or surgery depending on pregnancy status, eye disease, gland size, patient preference, and contraindications.

Case 5 - Cold, confused, and hypoventilating

Vignette: An elderly patient is somnolent with hypothermia, bradycardia, hyponatremia, hypercapnia, very high TSH, and low free T4.

Interpretation: Myxedema coma. Treat immediately with IV thyroid hormone, stress-dose glucocorticoids until adrenal insufficiency is excluded, passive warming, ventilatory support, and precipitant treatment.

Case 6 - Resistant hypertension and low potassium

Vignette: A patient requires four antihypertensives and has potassium 3.0 mEq/L with metabolic alkalosis. Renin is suppressed and aldosterone is elevated.

Interpretation: Primary aldosteronism. Confirm/localize as appropriate. Unilateral adenoma can be treated surgically; bilateral hyperplasia is treated with mineralocorticoid receptor antagonism.

Case 7 - Headache, sweating, palpitations

Vignette: A patient has episodic severe hypertension with pounding headache, diaphoresis, palpitations, tremor, and anxiety. Plasma metanephrines are elevated.

Interpretation: Pheochromocytoma/paraganglioma. Localize after biochemical confirmation. Preoperative alpha blockade is essential; beta blockade before alpha blockade can cause unopposed alpha vasoconstriction.

Case 8 - Weakness, bruising, diabetes, and striae

Vignette: A patient has proximal muscle weakness, easy bruising, new diabetes, infections, osteoporosis, and wide violaceous abdominal striae.

Interpretation: Cushing syndrome. Check exogenous steroid exposure first. If endogenous disease is suspected, screen with validated cortisol tests, then use ACTH to divide ACTH-dependent from ACTH-independent disease.

Case 9 - Amenorrhea and galactorrhea

Vignette: A woman has amenorrhea, infertility, galactorrhea, headaches, and markedly elevated prolactin.

Interpretation: Hyperprolactinemia suppresses GnRH and decreases LH/FSH, causing hypogonadism. Rule out pregnancy, hypothyroidism, renal disease, and medication effects. Prolactinoma usually responds to dopamine agonist therapy.

Case 10 - Enlarging shoes and new sleep apnea

Vignette: A patient reports larger ring and shoe sizes, jaw enlargement, sweating, OSA, hypertension, and glucose intolerance. IGF-1 is elevated.

Interpretation: Acromegaly. Confirm with failure of GH suppression during oral glucose testing when needed and image the pituitary. Transsphenoidal surgery is usually first-line when feasible.

Case 11 - Polyuria after pituitary surgery

Vignette: A postoperative pituitary patient makes 9 L/day of dilute urine with rising serum sodium and low urine osmolality.

Interpretation: Central diabetes insipidus. Replace free water, monitor sodium closely, and use desmopressin when appropriate. Exclude osmotic diuresis from hyperglycemia or mannitol.

Case 12 - Euvolemic hyponatremia after lung disease

Vignette: A patient with pneumonia has Na 119 mEq/L, low serum osmolality, urine osmolality 520 mOsm/kg, urine sodium 48 mEq/L, normal thyroid/adrenal function, and no edema.

Interpretation: SIADH. Treat the cause and restrict free water if stable. Severe neurologic symptoms require hypertonic saline with careful correction limits.

High-yield exam clues

Clues and links

Wide purple striae + proximal weakness + easy bruising	Cushing syndrome	More specific than nonspecific obesity or hypertension
HTN + hypokalemia + metabolic alkalosis	Primary aldosteronism	Aldosterone increases sodium reabsorption and K/H secretion
Headache + sweating + palpitations spells	Pheochromocytoma	Catecholamine spells can be fatal; alpha blockade first
Orbitopathy + pretibial myxedema	Graves disease	Extrathyroid findings are not typical of thyroiditis or toxic nodules
Painful tender thyroid after viral prodrome	Subacute thyroiditis	Low uptake; antithyroid drugs usually do not address mechanism
Delayed reflex relaxation	Hypothyroidism	Classic neuromuscular sign
Macroadenoma + bitemporal hemianopia	Pituitary mass compressing optic chiasm	Mass effect can occur with functional or nonfunctional adenomas
Very high prolactin	Prolactinoma	Mild elevation may be stalk effect, drugs, pregnancy, renal failure, hypothyroidism
No GH suppression after glucose	Acromegaly	Glucose normally suppresses GH; failure supports autonomous secretion
Low serum osm + concentrated urine + euvolemia	SIADH	ADH should be suppressed in hypotonicity
Dilute urine + hypernatremia + response to desmopressin	Central DI	Desmopressin replaces missing ADH

Common pitfalls

- Do not start insulin in DKA/HHS before checking potassium. If potassium is severely low, insulin can precipitate fatal arrhythmia.
- Do not give beta-blocker alone for pheochromocytoma. Alpha blockade must come first to avoid unopposed alpha vasoconstriction.
- Do not treat destructive thyroiditis with thionamides as if it were Graves disease; thionamides block new hormone synthesis, not release of preformed hormone.
- Do not use TSH alone when central hypothyroidism is possible. Free T4 carries the diagnosis in pituitary disease.
- Do not give levothyroxine to a patient with possible untreated adrenal insufficiency without considering glucocorticoid coverage.
- Do not assume normal potassium excludes primary aldosteronism. Many patients are normokalemic.
- Do not rapidly correct chronic hyponatremia. Severe symptoms require urgent therapy, but chronic adaptation makes overcorrection dangerous.
- Do not localize endocrine tumors before biochemical confirmation unless emergent mass effect demands imaging. Incidentalomas can mislead.
- Do not overlook medication causes: glucocorticoids, lithium, amiodarone, immune checkpoint inhibitors, dopamine antagonists, SSRIs, SGLT2 inhibitors, and antipsychotics can create endocrine patterns.

Practice questions

1. A patient with DKA has glucose 560 mg/dL, pH 7.12, bicarbonate 8 mEq/L, potassium 2.9 mEq/L. What is the immediate priority?

Answer: Replace potassium and hold insulin until potassium is safely above severe hypokalemia thresholds. Insulin would shift potassium intracellularly and can precipitate arrhythmia.

2. A patient has thyrotoxicosis, painful thyroid, high ESR, and low radioactive iodine uptake. Best disease-directed treatment?

Answer: Subacute thyroiditis: beta-blocker for symptoms plus NSAIDs or glucocorticoids for pain/inflammation; antithyroid drugs are not useful for the main mechanism.

3. A patient with resistant hypertension has aldosterone elevated and renin suppressed. What mechanism explains hypokalemia?

Answer: Aldosterone stimulates distal sodium reabsorption through ENaC with a lumen-negative potential t
potassium and hydrogen secretion.

4. A patient with pheochromocytoma is scheduled for surgery. Which medication sequence is safest?

Answer: Alpha blockade first, then beta blockade if needed for tachycardia.

5. A woman has amenorrhea, galactorrhea, and prolactin elevation. Why does prolactin cause infertility?

Answer: High prolactin inhibits GnRH, reducing LH and FSH pulsatility and causing hypogonadism/anovulation.

6. A postoperative pituitary patient has polyuria, urine osmolality 90 mOsm/kg, serum sodium 150 mEq/L, and urine osmolality rises after desmopressin. Diagnosis?

Answer: Central diabetes insipidus.

7. A patient with hyponatremia has low serum osmolality, urine osmolality 600 mOsm/kg, urine sodium 50 mEq/L, normal adrenal/thyroid function, and no edema. Diagnosis?

Answer: SIADH, after excluding adrenal insufficiency, hypothyroidism, renal failure, diuretics, and other mimics.

8. A patient has high TSH and low free T4 with bradycardia, constipation, and delayed reflex relaxation. Treatment?

Answer: Levothyroxine for overt primary hypothyroidism, with cautious dosing in older patients or coronary disease.

9. A patient has central hypothyroidism and secondary adrenal insufficiency. Which hormone replacement first?

Answer: Glucocorticoid replacement before thyroid hormone to avoid precipitating adrenal crisis.

10. A patient has high IGF-1 and GH fails to suppress after oral glucose. Most likely diagnosis and first-line localization?

Answer: Acromegaly from GH excess; pituitary MRI is used to localize an adenoma.

Graph-RAG extraction map

The following map is designed to help a graph-RAG pipeline extract durable nodes and edges. Nodes should store definitions, mechanisms, diagnostic anchors, management anchors, complications, and source contexts. Edges should store relationship type and rationale.

Node extraction schema

Diabetes mellitus	Endocrine/metabolic disease; glucose disorder	Type, criteria, insulin status, chronic complications, acute crises, medications	DM causes nephropathy/retinopathy/neuropat insulin deficiency causes DKA; insulin res associated with obesity/MASLD/CVD
DKA	Endocrine emergency; acid-base disorder	Ketosis, anion-gap acidosis, hyperglycemia/euglycemia, potassium shifts, treatment sequence	Insulin deficiency causes lipolysis/ketoge insulin therapy lowers potassium; infectio triggers DKA
HHS	Endocrine emergency; osmolar disorder	Hyperosmolality, dehydration, minimal ketosis, older T2DM pattern	Hyperglycemia causes osmotic diuresis; deh causes hyperosmolality/altered mental stat
Graves disease	Thyroid autoimmunity; hormone excess	TSH receptor antibodies, diffuse uptake, orbitopathy, pretibial myxedema	TSI stimulates TSH receptor; Graves causes hyperthyroidism; methimazole treats hormon synthesis
Thyroiditis	Thyroid inflammatory disorder	Painful/painless/postpartum, low uptake, transient phases	Follicular injury releases preformed hormo thyroiditis may progress to hypothyroid ph
Hypothyroidism	Thyroid hormone deficiency	Primary vs central, TSH/free T4 pattern, signs, treatment	Hashimoto causes primary hypothyroidism; p disease causes central hypothyroidism; levothyroxine treats hypothyroidism
Cushing syndrome	Adrenal/pituitary hormone excess	Cortisol excess, ACTH dependent/independent, screening tests	Cortisol excess causes diabetes/HTN/osteoporosis/infections; pitu adenoma causes Cushing disease
Pheochromocytoma	Adrenal medulla tumor; catecholamine excess	Metanephrines, spells, MEN2 link, alpha blockade	Catecholamines cause paroxysmal HTN; alpha blockade precedes beta blockade
Primary aldosteronism	Mineralocorticoid excess; secondary HTN	Aldosterone-renin ratio, hypokalemia, adrenal localization	Aldosterone causes Na retention/K wasting/alkalosis; adrenal adenoma treated adrenalectomy
Adrenal insufficiency	Hormone deficiency; endocrine emergency	Primary vs central, cortisol/ACTH/aldosterone/renin, crisis treatment	Adrenal cortex failure causes high ACTH + hyperkalemia; chronic steroids cause centr adrenal insufficiency
Prolactinoma	Pituitary adenoma; hormone excess	Prolactin level, hypogonadism, galactorrhea, mass effect	Prolactin inhibits GnRH; cabergoline treat prolactinoma
Acromegaly	Pituitary GH excess	IGF-1, failed GH suppression, cardiometabolic risks	GH excess increases IGF-1; acromegaly caus cardiomyopathy/OSA/diabetes
Central DI	Posterior pituitary water disorder	ADH deficiency, dilute urine, desmopressin response	ADH deficiency causes free water loss; desmopressin treats central DI
SIADH	Water retention disorder; hyponatremia syndrome	Euvolemic hypotonic hyponatremia, concentrated urine, causes	Excess ADH causes water retention; lung disease/CNS disease/drugs trigger SIADH

Suggested edge classes

- causes: Insulin deficiency causes ketogenesis; store biochemical pathway and expected labs.
- mimics: Thyroiditis mimics Graves disease; store discriminator such as uptake pattern.
- diagnosed_by: Acromegaly diagnosed by IGF-1 and GH suppression testing; store test limitations.
- treated_by: Central DI treated by desmopressin; store acute/chronic context and monitoring risks.
- sequence_warning: Beta blocker before alpha blockade is unsafe in pheochromocytoma.
- complicates: Diabetes complicates kidney, eye, nerve, and cardiovascular systems.
- primary_secondary_axis: Low cortisol + high ACTH indicates primary adrenal insufficiency.

Endocrine rapid-review checklist

- Identify the hormone axis: glucose/insulin, thyroid, adrenal, pituitary, ADH/water, calcium/PTH/vitamin D.
- Determine direction: hormone excess, hormone deficiency, receptor resistance, or inappropriate hormone timing.
- Localize: target gland primary problem, pituitary/hypothalamic secondary problem, ectopic source, medication effect, or destructive release.
- Screen safely: use the least misleading initial test for the axis and pretest probability.
- Stabilize emergencies before perfect localization: fluids, electrolytes, steroids, beta-blockade, insulin, ventilatory support, hypertonic saline, or dextrose as needed.

- Think medication and pregnancy: endocrine labs and treatments are altered by pregnancy, steroids, dop
lithium, amiodarone, iodine load, immune checkpoint inhibitors, SGLT2 inhibitors, and estrogen states.
- Follow complications longitudinally: endocrine diseases often kill through cardiovascular, renal, infectious, neurologic, and skeletal consequences rather than through the hormone number alone.

References used for clinical cross-checking

American Diabetes Association Standards of Care in Diabetes, including current recommendations for diagnosis, hospital glycemic care, and hyperglycemic crises. American Thyroid Association guideline resources for hyperthyroidism, hypothyroidism, thyroid nodules, thyroid cancer, and thyroid eye disease. Endocrine Society guideline resources for adrenal insufficiency, primary aldosteronism, Cushing syndrome, pheochromocytoma/paraganglioma, hyperprolactinemia, and pituitary disease. Uploaded Medvale source segments were used as coverage guides for diabetes, thyroid, adrenal, pituitary, DI, and SIADH topics.